A First Course in Abstract Algebra

Part 5: Ideals and Factor Rings

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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26 Homomorphisms and Factor Rings

Exercise 26.1: Homomorphisms of the product of integer rings

Plan. A homomorphism is determined by $u = \phi(1,0)$ and $v = \phi(0,1)$, since $\phi(m,n) = mu + nv$. The identities $(1,0)^2 = (1,0)$, $(0,1)^2 = (0,1)$ and $(1,0)(0,1) = 0$ force $u^2 = u$, $v^2 = v$, and $uv = 0$. An integer idempotent satisfies $t(t-1) = 0$, so each coordinate of u, v is 0 or 1. In each output coordinate, the pair of coefficients is therefore $(0,0)$, $(1,0)$, or $(0,1)$. Consequently each of the two coordinates of $\phi(m,n)$ can independently be 0, m, or n: there are $3^2 = 9$ maps. Conversely these nine maps preserve coordinatewise sums and products, so the list is exhaustive. Homomorphisms here are not required to preserve the unity of the codomain.

Exercise 26.2: A two-element subring

A subring isomorphic to $\mathbb{Z}_2$ must be $\{0, e\}$ with $e \neq 0$, $2e = 0$, and $e^2 = e$ modulo n. The additive cyclic group $\mathbb{Z}_n$ has an element of order two exactly when $n = 2m$, and then its only such element is $e = m$. Now $m^2 \equiv m \pmod{2m}$ is equivalent to $2 \mid (m-1)$. Thus m must be odd.

Conclusion. Exactly $n \equiv 2 \pmod 4$ work, with subring $\{0, n/2\}$. Its nonzero element acts as the identity within that subring.

Exercise 26.3: Ideals and quotients of the twelve-element ring

Every additive subgroup of $\mathbb{Z}_{12}$ is $\langle d \rangle$ for a divisor d of 12, and it is an ideal since multiplying a multiple of d by any residue still gives a multiple of d. Reduction modulo d has kernel $\langle d \rangle$, so

N	$\mathbb{Z}_{12}/N$
$\langle 1 \rangle = \mathbb{Z}_{12}$	$\{0\}$
$\langle 2 \rangle = \{0,2,4,6,8,10\}$	$\mathbb{Z}_2$
$\langle 3 \rangle = \{0,3,6,9\}$	$\mathbb{Z}_3$
$\langle 4 \rangle = \{0,4,8\}$	$\mathbb{Z}_4$
$\langle 6 \rangle = \{0,6\}$	$\mathbb{Z}_6$
$\langle 12 \rangle = \{0\}$	$\mathbb{Z}_{12}$

These are all the ideals because an ideal must first be an additive subgroup.

Exercise 26.4: Tables for an even-integer quotient

Write $A = 8\mathbb{Z}$, $B = 2 + 8\mathbb{Z}$, $C = 4 + 8\mathbb{Z}$, and $D = 6 + 8\mathbb{Z}$. Add or multiply representatives and then reduce modulo 8:

$+$	A	B	C	D
A	A	B	C	D
B	B	C	D	A
C	C	D	A	B
D	D	A	B	C

$\cdot$	A	B	C	D
A	A	A	A	A
B	A	C	A	C
C	A	A	A	A
D	A	C	A	C

For example $BD = 12 + 8\mathbb{Z} = C$. Every element squares to A or C, and no element is a multiplicative identity: $uC = A \neq C$ for all u. Since $\mathbb{Z}_4$ has an identity, these rings are not isomorphic, although their additive groups are isomorphic.

Exercise 26.5: Correcting isomorphism

The condition $\operatorname{Ker}\phi = \{0\}$ gives injectivity, but not surjectivity onto the declared codomain. An isomorphism $R \to R'$ is a *bijective* ring homomorphism. For instance inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ has zero kernel and is not an isomorphism onto $\mathbb{Q}$.

Exercise 26.6: The ideal definition

The definition is correct: an ideal is an additive subgroup N with $rn, nr \in N$ for every $r \in R, n \in N$. Both absorption conditions are needed in a noncommutative ring; they coincide when multiplication is commutative.

Exercise 26.7: Correcting the kernel

The kernel consists of *inputs*, not equations or outputs:

$$\operatorname{Ker}\phi = \{r \in R : \phi(r) = 0_{R'}\}.$$

For example the kernel of reduction $\mathbb{Z} \to \mathbb{Z}_5$ is $5\mathbb{Z}$, a subset of $\mathbb{Z}$.

Exercise 26.8: Differentiation

Differentiation preserves addition, but the product rule is $\delta(fg) = f'g + fg'$, rather than $f'g'$. Take $f = g = x$: $\delta(x^2) = 2x$ and $\delta(x)\delta(x) = 1$. Thus δ is not a ring homomorphism. Its kernel is the subring of constant functions (a differentiable function with derivative zero is constant). Example 26.12 shows this subring is not an ideal; this is another obstruction, since kernels of ring homomorphisms must be ideals.

Exercise 26.9: An image identity need not be the codomain identity

Let $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be $\phi(n) = (n, 0)$. Coordinatewise calculation gives $\phi(m + n) = \phi(m) + \phi(n)$ and $\phi(mn) = \phi(m)\phi(n)$. Yet $\phi(1) = (1, 0)$ is nonzero and differs from the codomain identity $(1, 1)$. It is the identity only on the image.

Exercise 26.10: True and false statements

(a) True. A homomorphism induces $R/\operatorname{Ker}\phi \cong \phi[R]$.

(b) False. Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ sends the ideal $\mathbb{Z}$ to $\mathbb{Z}$, which is not an ideal of $\mathbb{Q}$, since $(1/2)1 \notin \mathbb{Z}$.

(c) True. $\phi(r) = \phi(s)$ exactly when $r - s \in \operatorname{Ker}\phi$.

(d) False. $1 \in \mathbb{Q}$ but $\sqrt{2}\,1 \notin \mathbb{Q}$, so absorption in $\mathbb{R}$ fails.

(e) True. Additive closure is required, and ideal absorption includes closure under multiplication of its own elements.

(f) False. $\mathbb{Z} \subset \mathbb{Q}$ is the counterexample from (b).

(g) True. $(r+N)(s+N) = rs+N = sr+N$ if R is commutative.

(h) True. Reduction modulo four induces $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4$.

(i) True. If $1 \in N$, every $r = r1 \in N$; conversely $N = R$ contains 1.

(j) True. These are precisely the respective substructures allowing well-defined quotient operations.

Exercise 26.11: The extreme quotients

The ring R/R has just the one coset R and is the zero ring. The map $r + \{0\} \mapsto r$ identifies $R/\{0\}$ with R. Thus the two extreme quotients contain no new structure, although both are useful boundary cases in general theorems.

Exercise 26.12: A domain with a field quotient

Take the integral domain $\mathbb{Z}$ and its ideal $2\mathbb{Z}$. Reduction modulo two gives $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$, which is a field because its only nonzero element is its own inverse.

Exercise 26.13: A domain with zero divisors in a quotient

Again start with $\mathbb{Z}$, but quotient by $6\mathbb{Z}$. In $\mathbb{Z}/6\mathbb{Z}$, the cosets $2+6\mathbb{Z}$ and $3+6\mathbb{Z}$ are nonzero and their product is $6+6\mathbb{Z} = 0+6\mathbb{Z}$. Thus an integral domain can have a quotient with zero divisors.

Exercise 26.14: Removing zero divisors by a quotient

The ring $R = \mathbb{Z} \times \mathbb{Z}$ has $(1,0)(0,1) = 0$ with both factors nonzero. Projection $(a,b) \mapsto a$ has ideal kernel $N = \{0\} \times \mathbb{Z}$ and gives $R/N \cong \mathbb{Z}$, an integral domain.

Exercise 26.15: A subring that is not an ideal

The diagonal $D = \{(n,n) : n \in \mathbb{Z}\}$ is closed under differences and products and contains zero. But $(1,0)(1,1) = (1,0) \notin D$, so it is not an ideal of $\mathbb{Z} \times \mathbb{Z}$.

Exercise 26.16: Representatives and quotient elements

(a) The student has used r, s for elements of R/N and then written $rs - sr \in N$, which should concern elements of R. Quotient elements are cosets, so the types of objects have been confused. **(b)** A correct start is: for all $r, s \in R$, commutativity gives $(r+N)(s+N) = (s+N)(r+N)$. **(c)** That equality is equivalent to $rs+N = sr+N$, which is equivalent to $rs - sr \in N$. Reading these equivalences in either direction proves both implications. Notice that $rs = sr$ in R itself is not required.

Exercise 26.17: A quadratic integer ring and matrices

Plan. Match the coefficients of 1 and $\sqrt{2}$ with matrix entries. If $u = a + b\sqrt{2}$ and $v = c + d\sqrt{2}$, then

$$u - v = (a-c) + (b-d)\sqrt{2}, \qquad uv = (ac+2bd) + (ad+bc)\sqrt{2}.$$

These formulas show R is a nonempty subring of $\mathbb{R}$. Set $M(a,b) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$. Matrix subtraction and multiplication give

$$M(a,b) - M(c,d) = M(a-c, b-d), \quad M(a,b)M(c,d) = M(ac+2bd, ad+bc),$$

so R' is a subring of $M_2(\mathbb{Z})$ and $\phi(a+b\sqrt{2}) = M(a,b)$ preserves the operations. Coefficients in $a+b\sqrt{2}$ are unique: otherwise irrationality of $\sqrt{2}$ would fail. Hence the map is well defined; its kernel is zero by its entries, and every displayed matrix is an image. Thus it is a ring isomorphism.

Exercise 26.18: Homomorphisms from a field

Let $\phi : F \to R$ have a nonzero kernel element a. For any $x \in F$, write $x = (xa^{-1})a$. Then $\phi(x) = \phi(xa^{-1})\phi(a) = 0$, so ϕ is the zero map. If no such a exists, the kernel is zero, and ϕ is injective. These are the only possibilities.

Exercise 26.19: Composing ring homomorphisms

For $a, b \in R$, apply the defining identities twice:

$$\psi\phi(a+b) = \psi(\phi(a) + \phi(b)) = \psi\phi(a) + \psi\phi(b),$$
$$\psi\phi(ab) = \psi(\phi(a)\phi(b)) = \psi\phi(a)\,\psi\phi(b).$$

Thus $\psi \circ \phi : R \to R''$ preserves both operations and is a ring homomorphism.

Exercise 26.20: Frobenius

In a commutative ring the binomial theorem applies. For prime p, each $\binom{p}{k}$ with $0 < k < p$ is divisible by p, so characteristic p gives

$$(a+b)^p = a^p + b^p.$$

Commutativity also gives $(ab)^p = a^p b^p$. Therefore $a \mapsto a^p$ preserves addition and multiplication and is a ring homomorphism. Commutativity is essential to these expansions.

Exercise 26.21: The image of the identity in a ring without zero divisors

Put $e = \phi(1)$. It satisfies $e^2 = e$. It is nonzero, because otherwise $\phi(r) = \phi(1r) = e\phi(r) = 0$ for every r, contrary to the nonzero-image assumption. For any $y \in R'$, we have $e(ey - y) = 0$. Since $e \neq 0$ and R' has no zero divisors, $ey = y$. Likewise $(ye - y)e = 0$ gives $ye = y$. Thus e is a two-sided identity for all of R', even when ϕ is not onto.

Exercise 26.22: Images and inverse images of ideals

(a) The image $\phi[N]$ is an additive subgroup. For $\phi(r) \in \phi[R]$ and $\phi(n) \in \phi[N]$, both products equal images of rn or nr in N, so absorption holds inside $\phi[R]$. **(b)** Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with $N = \mathbb{Z}$ fails absorption by $1/2$, giving the required counterexample. **(c)** Let $I = \phi^{-1}[N']$. Then $0 \in I$, and for $a, b \in I$, $\phi(a-b) = \phi(a) - \phi(b) \in N'$. For $r \in R$, both $\phi(ra) = \phi(r)\phi(a)$ and $\phi(ar) = \phi(a)\phi(r)$ lie in N'. Thus I is an ideal. This uses only absorption by elements in $\phi[R]$, so either stated choice of ambient ring for N' is sufficient.

Exercise 26.23: Polynomials vanishing on a set

Let N_S be the polynomials vanishing at each point of $S \subset F^n$. The zero polynomial belongs to N_S. If $f, g \in N_S$, then $(f-g)(a) = f(a) - g(a) = 0$ for every $a \in S$. If h is any polynomial, $(hf)(a) = h(a)f(a) = 0$ and similarly for fh. Thus the subring test and absorption show that N_S is an ideal. If S is empty, the condition is vacuous and $N_S = F[x_1, \ldots, x_n]$, still an ideal.

Exercise 26.24: Quotients of a field

An ideal N of a field F is either zero or all of F: if $0 \neq a \in N$, then $1 = a^{-1}a \in N$, and hence every $x = x1 \in N$. In the two cases, F/N is respectively isomorphic to F or to the one-element zero ring.

Exercise 26.25: Unity in a proper quotient

The coset $1 + N$ multiplies any $r + N$ to $r + N$ on either side. It is nonzero because $1 \in N$ would force $N = R$, contrary to the assumption. Thus a proper quotient of a ring with unity again has a nonzero unity.

Exercise 26.26: An annihilator ideal

Let $I_a = \{x : ax = 0\}$. It contains zero. If $x, y \in I_a$, then $a(x - y) = 0 - 0 = 0$, so it is an additive subgroup. For $r \in R$, commutativity gives $a(rx) = r(ax) = 0$, hence $rx = xr \in I_a$. Thus I_a is an ideal. Without commutativity, a one-sided statement is automatic but both sides need not follow.

Exercise 26.27: Intersections of ideals

For a family (I_j) of ideals, zero belongs to every I_j. If a, b belong to their intersection, $a - b \in I_j$ for every j, hence lies in the intersection. Likewise ra and ar belong to every I_j for every $r \in R$. This proves the ideal axioms. The intersection of an empty family is understood as R, also an ideal.

Exercise 26.28: Induced quotient homomorphism

Define $\overline{\phi}(r + N) = \phi(r) + N'$. If $r + N = s + N$, then $r - s \in N$ and $\phi(r) - \phi(s) \in \phi[N] \subset N'$, so the output cosets agree. Thus the formula is well defined. Moreover

$$\overline{\phi}((r + N) + (s + N)) = \phi(r + s) + N' = \overline{\phi}(r + N) + \overline{\phi}(s + N),$$

and the identical calculation using products proves multiplicativity. No surjectivity assumption is needed.

Exercise 26.29: Images of units under a surjection

Because ϕ is onto, $\phi(1)$ acts as a two-sided identity on every element of R': write it as $\phi(r)$ and use $1r = r1 = r$. Also $\phi(1) \neq 0$ since $R' \neq \{0\}$. If $uv = vu = 1$ in R, applying ϕ gives $\phi(u)\phi(v) = \phi(v)\phi(u) = 1_{R'}$. Thus $\phi(u)$ is a unit with inverse $\phi(v)$.

Exercise 26.30: The nilradical

Let N consist of nilpotent elements in a commutative ring. It contains zero. If $a^m = 0$ and $b^n = 0$, expand $(a-b)^{m+n-1}$. Each term contains either at least m factors a or at least n factors b, so every term is zero. Thus $a - b \in N$. For $r \in R$, $(ra)^m = r^m a^m = 0$, so $ra = ar \in N$. These facts prove that N is an ideal. Commutativity is used in both power calculations.

Exercise 26.31: Three nilradicals

In $\mathbb{Z}_{12}$ a nilpotent residue must be divisible by both primes $2, 3$, since reducing a vanishing power modulo either prime gives a vanishing element in a field. Conversely $6^2 \equiv 0 \pmod{12}$. Thus the nilradical is $\{0, 6\} = \langle 6 \rangle$. In $\mathbb{Z}$, a vanishing integer power forces the integer to be zero, so the nilradical is $\{0\}$. In $\mathbb{Z}_{32}$, precisely the even residues are nilpotent: $(2k)^5 \equiv 0$, while an odd residue is a unit and cannot have a zero power. Thus the nilradical is $\langle 2 \rangle$.

Exercise 26.32: Quotienting by all nilpotents

Suppose $(a + N)^m = N$ in R/N. Then $a^m \in N$, and because N is the nilradical, $(a^m)^n = 0$ for some n. Hence $a^{mn} = 0$, so $a \in N$ and $a + N = N$. Thus the quotient has no nonzero nilpotent elements.

Exercise 26.33: Lifting nilpotence

For any $a \in R$, the hypothesis that every coset is nilpotent gives $(a + N)^m = N$, hence $a^m \in N$. Every element of N is nilpotent, so $(a^m)^n = 0$ for some n. Thus every $a \in R$ is nilpotent, which says the nilradical is all of R. The exponents may depend on a.

Exercise 26.34: The radical of an ideal

Let $\sqrt{N} = \{a : a^m \in N \text{ for some } m \geq 1\}$. It contains zero. If $a^m, b^n \in N$, expand $(a-b)^{m+n-1}$; each term has a factor a^m or b^n, so each term belongs to N. This puts $a - b$ in $\sqrt{N}$. For any $r \in R$, $(ra)^m = r^m a^m \in N$, so $ra = ar \in \sqrt{N}$. Therefore $\sqrt{N}$ is an ideal. This proof does not require all elements of N themselves to be nilpotent.

Exercise 26.35: A radical can grow or remain the same

(a) In $\mathbb{Z}$, take $N = 4\mathbb{Z}$. Then $2^2 \in 4\mathbb{Z}$ but $2 \notin 4\mathbb{Z}$, so $\sqrt{N} \neq N$ (in fact $\sqrt{N} = 2\mathbb{Z}$). **(b)** Take $N = 2\mathbb{Z}$. If a^m is even then a is even, so $\sqrt{2\mathbb{Z}} = 2\mathbb{Z}$. Both examples use proper ideals.

Exercise 26.36: Radical versus quotient nilradical

An element of the nilradical of R/N is a *coset* $a + N$ such that $(a + N)^m = N$, equivalently $a^m \in N$. Hence

$$\mathrm{Nil}(R/N) = \{a + N : a \in \sqrt{N}\} = \sqrt{N}/N.$$

It is not literally $\sqrt{N}$, whose elements lie in R rather than R/N.

Exercise 26.37: Complex numbers as real matrices

Set $M(a,b) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Direct multiplication gives

$$M(a,b)M(c,d) = M(ac - bd, ad + bc),$$

which matches $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. Addition also matches entry by entry. The map is injective because $M(a,b) = 0$ forces $a = b = 0$, and it is onto its image by definition. Therefore $\mathbb{C} \cong \phi[\mathbb{C}] \subset M_2(\mathbb{R})$. The sign convention corresponds to multiplication on coordinates $(x, -y)$ instead of (x, y) and is consistent.

Exercise 26.38: A ring version of the regular representation

(a) Left multiplication λ_a is additive because $a(x + y) = ax + ay$, so it is an endomorphism of $(R, +)$.
(b) Endomorphisms are added pointwise and multiplied by composition. Thus

$$\lambda_a - \lambda_b = \lambda_{a-b}, \qquad \lambda_a \circ \lambda_b = \lambda_{ab}.$$

Together with $\lambda_0 = 0$, these identities show that $R' = \{\lambda_a : a \in R\}$ is a subring. **(c)** The map $a \mapsto \lambda_a$ is a surjective ring homomorphism onto R'. If $\lambda_a = \lambda_b$, evaluate at 1 to obtain $a = b$. Thus it is injective and $R \cong R'$. Unity is what makes this final injectivity argument work.

27 Prime and Maximal Ideals

Exercise 27.1: Prime and maximal ideals in a six-element ring

Every ideal is $\langle d\rangle$ for $d \mid 6$, and $\mathbb{Z}_6/\langle d\rangle \cong \mathbb{Z}_d$. A proper ideal is prime exactly when its quotient is an integral domain, and maximal exactly when its quotient is a field. Among the divisors $1, 2, 3, 6$, exactly $2, 3$ are prime. Thus both lists are $\langle 2\rangle = \{0, 2, 4\}$ and $\langle 3\rangle = \{0, 3\}$.

Exercise 27.2: Prime and maximal ideals in a twelve-element ring

For $d \mid 12$, reduction modulo d gives $\mathbb{Z}_{12}/\langle d\rangle \cong \mathbb{Z}_d$. The quotient is a domain, equivalently a field in this finite case, exactly when $d = 2$ or 3. Hence the prime ideals and the maximal ideals are both $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$ and $\langle 3\rangle = \{0, 3, 6, 9\}$.

Exercise 27.3: A product of two two-element fields

In a product $A \times B$ with identities, every ideal is $I \times J$: multiply (a, b) by $(1, 0)$ and $(0, 1)$ to separate the coordinates. A domain quotient cannot retain both nonzero factors, since $(1, 0)(0, 1) = 0$. Thus the prime and maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are exactly $\{0\} \times \mathbb{Z}_2$ and $\mathbb{Z}_2 \times \{0\}$. Each quotient is $\mathbb{Z}_2$.

Exercise 27.4: A product involving a four-element ring

The same product argument shows that a prime ideal must be all of one factor times a prime ideal in the other. In $\mathbb{Z}_2$ the only prime ideal is zero, and in $\mathbb{Z}_4$ it is $\langle 2\rangle = \{0, 2\}$. Therefore the only prime ideals are

$$\{0\} \times \mathbb{Z}_4, \qquad \mathbb{Z}_2 \times \{0, 2\}.$$

Both quotients are $\mathbb{Z}_2$, so these are also exactly the maximal ideals.

Exercise 27.5: Quadratic quotients over the three-element field

A quotient by a nonconstant polynomial is a field exactly when that polynomial is irreducible. A quadratic over a field is irreducible exactly when it has no root. In $\mathbb{Z}_3$, the possible squares are $0, 1$. The equation $x^2 = -c$ has no solution exactly when $-c = 2$, or $c = 1$. Thus only $c = 1$ works.

Exercise 27.6: A cubic family with fixed quadratic term

A cubic over a field is reducible exactly when it has a root. At $x = 0, 1, 2$, the values of $x^3 + x^2$ modulo three are $0, 2, 0$. Thus $x^3 + x^2 + c$ has a root if $c = 0$ or 1, but for $c = 2$ its values are $2, 1, 2$, all nonzero. The required value is $c = 2$.

Exercise 27.7: A cubic family with a variable coefficient

Evaluate $x^3 + cx^2 + 1$ at $0, 1, 2$. For $c = 0$ the value at 2 is zero; for $c = 1$ the value at 1 is zero; for $c = 2$ the three values are $1, 1, 2$, all nonzero. Thus exactly $c = 2$ makes the cubic irreducible and the quotient a field.

Exercise 27.8: Quadratics with a linear term over five elements

Since the characteristic is not two, $x^2 + x + c$ has a root exactly when the discriminant $1 - 4c$ is a square. The squares in $\mathbb{Z}_5$ are $0, 1, 4$. For $c = 0, 1, 2, 3, 4$, the discriminants are respectively $1, 2, 3, 4, 0$. Thus the nonsquare cases are $c = 1, 2$, and precisely those quotients are fields.

Exercise 27.9: Quadratics with constant term one

For $x^2 + cx + 1$, the discriminant is $c^2 - 4$. As c runs through $0, 1, 2, 3, 4$ in $\mathbb{Z}_5$, the discriminants are $1, 2, 0, 0, 2$. Nonsquares occur for $c = 1, 4$, so exactly those values give fields.

Exercise 27.10: Correcting maximal ideal

A maximal ideal M is a *proper* ideal of R such that the only ideals containing it are M and R. Merely saying that no other ideal contains it would make R the only candidate, because R contains every ideal.

Exercise 27.11: Correcting prime ideal

In a commutative ring, a prime ideal P is a proper ideal such that $ab \in P$ implies $a \in P$ or $b \in P$ for all $a, b \in R$. It need not be generated by a prime number; for example $(x) \subset \mathbb{Q}[x]$ is prime because its quotient is $\mathbb{Q}$.

Exercise 27.12: Prime field

The definition is correct: a prime field has no proper subfields. Equivalently it equals the subfield generated by its unity. Up to isomorphism these fields are $\mathbb{Q}$ and $\mathbb{Z}_p$ for primes p.

Exercise 27.13: Principal ideal

The definition is correct. In a commutative ring with unity, the smallest ideal containing a is $(a) = aR = \{ar : r \in R\}$. It contains $a = a1$, is closed under differences and absorption, and every ideal containing a contains all its multiples.

Exercise 27.14: True and false statements

(a) False. $(0) \subset \mathbb{Z}$ is prime but not maximal, since $\mathbb{Z}$ is a domain and not a field.

(b) True. The quotient by a maximal ideal is a field and hence a domain.

(c) True. Every subfield of $\mathbb{Q}$ contains 1, all integers, and all nonzero integer inverses, hence all rationals.

(d) False. The prime subfield of $\mathbb{C}$ is $\mathbb{Q}$.

(e) True. The unity generates a copy of $\mathbb{Q}$ in characteristic zero or of $\mathbb{Z}_p$ in characteristic p.

(f) True. $\mathbb{Z}_2 \times \mathbb{Z}_2$ has zero divisors and contains the diagonal field $\{(0,0),(1,1)\}$.

(g) True. Send m/n to $(m1)(n1)^{-1}$; characteristic zero ensures no nonzero integer maps to zero.

(h) False. $(x^2) \subset F[x]$ is not prime: x^2 belongs but x does not.

(i) True. Choose a nonzero element of smallest degree and use the division algorithm; include the zero ideal separately.

(j) False. (x^2) is principal but its quotient has a nonzero nilpotent element $x + (x^2)$, so is not a field.

Exercise 27.15: A maximal ideal of an integer product

Let $M = 2\mathbb{Z} \times \mathbb{Z}$. Projection onto the first coordinate followed by reduction modulo two is onto $\mathbb{Z}_2$ and has kernel M. Thus $(\mathbb{Z} \times \mathbb{Z})/M \cong \mathbb{Z}_2$, a field, and M is maximal.

Exercise 27.16: A prime ideal that is not maximal

Let $P = \{0\} \times \mathbb{Z}$. Projection onto the first coordinate gives quotient $\mathbb{Z}$, a domain but not a field. Thus P is prime and not maximal. Explicitly $P \subsetneq 2\mathbb{Z} \times \mathbb{Z} \subsetneq \mathbb{Z} \times \mathbb{Z}$.

Exercise 27.17: An ideal that is not prime

Let $I = 4\mathbb{Z} \times \mathbb{Z}$. It is proper and nonzero. Neither factor $(2,0)$ of $(2,0)(2,0) = (4,0)$ lies in I, but their product does. Hence I is not prime.

Exercise 27.18: A reducible rational quadratic quotient

Since $x^2 - 5x + 6 = (x-2)(x-3)$, the defining polynomial is reducible over $\mathbb{Q}$. In the quotient, the two linear-factor cosets are nonzero (their degrees are too small to be multiples of the quadratic), and their product is zero. Thus the quotient is not a field.

Exercise 27.19: An irreducible rational quadratic quotient

The discriminant of $x^2 - 6x + 6$ is $36 - 24 = 12$, whose square root $2\sqrt{3}$ is irrational. Thus the quadratic has no rational root and is irreducible over $\mathbb{Q}$. Its generated ideal is maximal, so the quotient is a field, isomorphic to $\mathbb{Q}(\sqrt{3})$ by evaluating at $3 + \sqrt{3}$.

Exercise 27.20: Synopsis: maximality produces a field quotient

If M is maximal and $a \notin M$, the ideal $M + (a)$ must be R, so $1 = m + ra$ for some $m \in M, r \in R$. Passing to cosets gives $(r + M)(a + M) = 1 + M$, so every nonzero coset has an inverse and R/M is a field.

Exercise 27.21: Synopsis: a field quotient forces maximality

If R/M is a field and I is an ideal properly containing M, choose $a \in I \setminus M$; an inverse of $a + M$ gives $ra - 1 \in M$ for some $r \in R$, so $1 \in I$ and $I = R$. Thus M has no proper intermediate ideal and is maximal.

Exercise 27.22: Synopsis: ideals in a polynomial ring

For a nonzero ideal $I \subset F[x]$, choose $g \in I$ of smallest degree. Dividing any $f \in I$ by g gives a remainder in I of smaller degree, hence zero; therefore $I = (g)$. The zero ideal is (0).

Exercise 27.23: Synopsis: maximal polynomial ideals are irreducible

If (p) is maximal but $p = fg$ with both factors of positive degree, then $(p) \subsetneq (f) \subsetneq F[x]$, a contradiction. Hence a nonconstant generator of a maximal ideal must be irreducible.

Exercise 27.24: Finite prime quotients

If P is prime, R/P is an integral domain and is finite because R is finite. For any nonzero a in a finite domain, the multiplication map $x \mapsto ax$ is injective by cancellation, hence surjective. It therefore takes some x to 1, giving an inverse for a. Thus R/P is a field, and P is maximal.

Exercise 27.25: Different cyclic-ring subrings

It is possible when the subrings need not share the ambient unity. In $R = \mathbb{Z}_2 \times \mathbb{Z}_3$, the subrings $\mathbb{Z}_2 \times \{0\}$ and $\{0\} \times \mathbb{Z}_3$ are isomorphic to $\mathbb{Z}_2$ and $\mathbb{Z}_3$, respectively. Their identities are $(1, 0)$ and $(0, 1)$, not the ambient $(1, 1)$. This distinction avoids incorrectly equating their characteristics with that of R.

Exercise 27.26: Two different prime-field subrings

The same example $\mathbb{Z}_p \times \mathbb{Z}_q$, for distinct primes p, q, contains the field subrings $\mathbb{Z}_p \times \{0\}$ and $\{0\} \times \mathbb{Z}_q$. Each is a field under the inherited operations, with its own identity. Thus the answer is yes under the book's subring convention.

Exercise 27.27: The domain case

In an integral domain, a nonzero idempotent e satisfies $e(e - 1) = 0$, so $e = 1$. Therefore the identity of any field subring is the ambient identity. A copy of $\mathbb{Z}_p$ forces the ambient characteristic to be p, and a copy of $\mathbb{Z}_q$ forces it to be q. Since $p \neq q$ is impossible, no such domain exists. Equivalently distinct nonzero orthogonal identities would create zero divisors.

Exercise 27.28: Direct proof that maximal ideals are prime

Suppose $ab \in M$ and $a \notin M$. The ideal $M + (a)$ properly contains M, so maximality makes it R. Hence $1 = ra + m$ for some $r \in R, m \in M$. Multiply by b:

$$b = rab + mb \in M,$$

since both terms belong to M. This proves $a \in M$ or $b \in M$, and M is proper by definition, so it is prime.

Exercise 27.29: Maximal ideals and simple quotient rings

Ideals J containing N correspond to ideals J/N of R/N: inverse image under the quotient map reverses this assignment. To see the image direction, multiplication by $r + N$ sends $j + N$ to $rj + N \in J/N$; the inverse-image direction follows from the ideal test. Therefore the only ideals of R/N are zero and the whole quotient exactly when the only ideals containing N are N and R. Requiring a nontrivial quotient is equivalent to $N \neq R$. These statements say precisely that N is maximal if and only if R/N is simple.

Exercise 27.30: Nonzero prime ideals of a one-variable polynomial ring

Write the proper nonzero prime ideal as (p), using the principal-ideal theorem for $F[x]$. If $p = fg$ with $\deg f, \deg g > 0$, primality puts f or g in (p), impossible by degree. Thus p is irreducible. The quotient by an irreducible polynomial over a field is a field, so (p) is maximal.

Exercise 27.31: Divisibility and membership

By definition, $(f) = \{hf : h \in F[x]\}$. Thus $g \in (f)$ means exactly that $g = hf$ for some polynomial h, which is the statement $f \mid g$. This includes $f = 0$: membership and divisibility both then require $g = 0$.

Exercise 27.32: The ideal generated by two polynomials

The set $N = (f, g)$ contains zero and is closed under subtraction:

$$(rf + sg) - (r'f + s'g) = (r - r')f + (s - s')g.$$

Multiplying by any $h \in F[x]$ gives $(hr)f + (hs)g$, again in N, so it is an ideal. If both f, g were irreducible and had distinct degrees, their only common divisors would be constants: a nonconstant common divisor of an irreducible is an associate of it, forcing equal degrees. The Euclidean algorithm then gives $uf + vg = 1$, which would make $N = F[x]$. This contradicts the stated hypothesis; thus they cannot both be irreducible.

Exercise 27.33: A one-variable Nullstellensatz

Plan. Replace the finitely generated ideal by a single greatest common divisor. Assume the Fundamental Theorem of Algebra. Write $(f_1, \ldots, f_r) = (h)$ using the principal-ideal theorem. If

h is a nonzero constant, the ideal is the whole ring and the assertion is immediate. If $h = 0$, all f_i vanish everywhere; then g vanishes everywhere, so it is the zero polynomial and belongs to the ideal. Otherwise factor $h = c\prod_j (x - a_j)^{e_j}$. Its zeros are exactly the common zeros of the f_i, since h divides every f_i and is their polynomial linear combination. Each a_j is therefore a zero of g. Taking $m \geq \max_j e_j$ makes $h \mid g^m$, so $g^m \in (f_1, \ldots, f_r)$. Conversely assume the stated Nullstellensatz. If a nonconstant f had no complex zero, the condition on $g = 1$ would be vacuously true. It would follow that $1 = 1^m \in (f)$, impossible since a nonconstant polynomial cannot divide 1. Thus every nonconstant complex polynomial has a zero.

Exercise 27.34: The sum of two ideals

(a) Zero is $0 + 0 \in A + B$. Subtracting $a + b$ and $a' + b'$ yields $(a - a') + (b - b') \in A + B$. For $r \in R$, both $r(a + b) = ra + rb$ and $(a + b)r = ar + br$ lie in $A + B$. Thus this is an ideal even if R is noncommutative. **(b)** Since $0 \in A \cap B$, write $a = a + 0$ and $b = 0 + b$. This proves $A \subset A + B$ and $B \subset A + B$.

Exercise 27.35: The product of two ideals

(a) The zero sum can be represented as $0 \cdot 0$. Sums of finite sums of products remain such sums; a negative is obtained by replacing a_i by $-a_i$. Also

$$r\sum_i a_i b_i = \sum_i (ra_i)b_i, \qquad (\sum_i a_i b_i)r = \sum_i a_i(b_i r),$$

and the changed factors stay in the appropriate ideals. Hence AB is an ideal. **(b)** Each $a_i b_i$ lies in A by right absorption and in B by left absorption. Both ideals are closed under finite sums, so $AB \subset A \cap B$.

Exercise 27.36: The ideal quotient

Let $I = A : B = \{r : rb \in A \text{ for every } b \in B\}$. Zero belongs to I. If $r, s \in I$, then $(r-s)b = rb-sb \in A$ for every $b \in B$, so $r - s \in I$. If $t \in R$, then $(tr)b = t(rb) \in A$. Thus $tr \in I$, and commutativity gives the other side too. Therefore $A : B$ is an ideal.

Exercise 27.37: A one-sided matrix ideal

Let S be the matrices with second row zero. Differences and products of such matrices still have second row zero, so S is a subring. For any $B \in M_2(F)$, the second row of AB is zero whenever $A \in S$, so S is a right ideal. Using matrix units, $E_{11} \in S$ but $E_{21}E_{11} = E_{21} \notin S$. Therefore multiplication on the left by an arbitrary ambient matrix need not stay in S, and S is not a left ideal.

Exercise 27.38: Simplicity of a matrix ring

Let I be a nonzero two-sided ideal in $M_2(\mathbb{Z}_2)$, and choose $A \in I$ with nonzero entry $a_{ij} = 1$. For any $r, s \in \{1, 2\}$,

$$E_{ri}AE_{js} = a_{ij}E_{rs} = E_{rs} \in I.$$

Thus I contains all four matrix units. Every binary matrix is the sum of the matrix units corresponding to its nonzero entries, so $I = M_2(\mathbb{Z}_2)$. Hence the only two-sided ideals are zero and the whole ring, proving simplicity.

28 Gröbner Bases for Ideals

Exercise 28.1: Decreasing lexicographic order, 1

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-3x^3 + 7x^2y^2z - 5x^2yz^3 + 2xy^3z^5.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.2: Decreasing lexicographic order, 2

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-4x + 5y^3z^3 + 3y^2z^5 - 8z^7.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.3: Decreasing lexicographic order, 3

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$2x^2yz^2 - 2xy^2z^2 - 7x + 3y + 10z^3.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.4: Decreasing lexicographic order, 4

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-8xy - 4xz + 3yz^3 + 2yz + 38.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.5: Reversed variable priority, 1

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 1 becomes

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

Exercise 28.6: Reversed variable priority, 2

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 2 becomes

$$-8z^7 + 3y^2z^5 + 5y^3z^3 - 4x.$$

Exercise 28.7: Reversed variable priority, 3

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 3 becomes

$$10z^3 - 2xy^2z^2 + 2x^2yz^2 + 3y - 7x.$$

Exercise 28.8: Reversed variable priority, 4

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 4 becomes

$$3yz^3 + 2yz - 4xz - 8xy + 38.$$

Exercise 28.9: The first twenty degree-lexicographic monomials

First sort by total degree, then by lexicographic order with $z < y < x$. There are $1, 3, 6, 10$ monomials of degrees $0, 1, 2, 3$, respectively, so these are exactly the first twenty:

$$1;$$
$$z,\ y,\ x;$$
$$z^2,\ yz,\ y^2,\ xz,\ xy,\ x^2;$$
$$z^3,\ yz^2,\ y^2z,\ y^3,\ xz^2,\ xyz,\ xy^2,\ x^2z,\ x^2y,\ x^3.$$

For instance $y^3 < xz^2$ despite the larger y exponent, because the latter has a positive x exponent.

Exercise 28.10: Decreasing degree-lexicographic order, 1

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.11: Decreasing degree-lexicographic order, 2

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3y^2z^5 - 8z^7 + 5y^3z^3 - 4x.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.12: Decreasing degree-lexicographic order, 3

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2x^2yz^2 - 2xy^2z^2 + 10z^3 - 7x + 3y.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.13: Decreasing degree-lexicographic order, 4

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3yz^3 - 8xy - 4xz + 2yz + 38.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.14: A one-step reduction of the largest leading monomial

The largest leading monomial is x^2y, in $f_2 = x^2y + 4xy$. It is divisible by the leading monomial xy of $f_3 = xy - y^2$. Replace

$$f_2 \longmapsto f_2 - xf_3 = xy^2 + 4xy.$$

The new basis is $\{xy^2 - 2x,\ xy^2 + 4xy,\ xy - y^2\}$. Its maximum monomial is xy^2, smaller than x^2y. The ideal is unchanged because the replacement can be reversed by adding xf_3.

Exercise 28.15: Reducing a mixed cubic basis

The first polynomial has leading monomial xy; the third, $x - y^4$, has leading monomial x. Replace the first by

$$(xy + y^3) - y(x - y^4) = y^5 + y^3.$$

The resulting basis $\{y^5 + y^3,\ y^3 + z,\ x - y^4\}$ generates the same ideal. Its largest monomial is now x, whereas the previous largest was xy. In pure lex order a monomial involving x exceeds every monomial involving only y, z, regardless of degree.

Exercise 28.16: When the maximum cannot be reduced in one step

The leading monomials are xyz, x^3, and x^2yz^3. The largest is x^3. Neither xyz nor x^2yz^3 divides x^3, because each contains positive y, z exponents. Thus no division step using another basis element can cancel that largest monomial. One can reduce the third polynomial using the first, but the unchanged x^3 still remains the maximum. Consequently a single-step reduction that strictly decreases the maximum monomial of the entire basis is impossible.

Exercise 28.17: A reduction involving only two variables

The largest leading monomial is y^3z^2, divisible by y^2z^2, the leading monomial of the third polynomial. Replace

$$(y^3z^2 - 2z) - y(y^2z^2 + 3) = -3y - 2z.$$

Thus $\{y^2z^3 + 3,\ -3y - 2z,\ y^2z^2 + 3\}$ generates the same ideal. The maximum has dropped from y^3z^2 to y^2z^3.

Exercise 28.18: A linear Groebner basis

Name the three displayed generators f_1, f_2, f_3 in their source order. Ordinary elimination produces

$$\begin{aligned} g_1 &= 3f_1 - f_3 = 2w + 11z - 4, \\ g_2 &= -7f_1 + 2f_2 + 3f_3 = 4x - 29z + 14, \\ g_3 &= -5f_1 + 2f_2 + f_3 = 4y - 23z + 18. \end{aligned}$$

The coefficient transformation has nonzero determinant -8, so it is invertible over $\mathbb{R}$: both triples generate the same ideal. The leading monomials are w, x, y, pairwise relatively prime. Their S-polynomials therefore reduce to zero, so $\{g_1, g_2, g_3\}$ is a Groebner basis. Equivalently this is row reduction of a linear system, with z free.

Exercise 28.19: An inconsistent linear ideal

Let the source generators be f_1, f_2, f_3. Subtract twice the first from the second to obtain $6x - 5y + 1$. Subtract the first from the third to obtain $-6x + 5y - 3$. Adding gives

$$f_2 + f_3 - 3f_1 = -2.$$

Thus the ideal contains the nonzero constant -2, hence also 1. It is the whole polynomial ring, and $\{1\}$ is a Groebner basis. Its algebraic variety is empty.

Exercise 28.20: A one-variable common divisor

Let $f = x^4 + x^3 - 3x^2 - 4x - 4$ and $g = x^3 + x^2 - 4x - 4$. The first division gives

$$f - xg = x^2 - 4, \qquad g = (x + 1)(x^2 - 4).$$

Therefore $(f, g) = (x^2 - 4)$. A single generator of a one-variable ideal is a Groebner basis: the leading term of every multiple is divisible by the generator's leading term. The answer is $\{x^2 - 4\}$.

Exercise 28.21: Three one-variable generators

For the first two generators f_1, f_2, division gives

$$f_1 - (x - 3)f_2 = 6(x^2 - 3x + 2), \qquad f_2 = (x + 2)(x^2 - 3x + 2).$$

Their gcd is $h = (x - 1)(x - 2)$. The third generator is $f_3 = (x - 1)^2(x + 2)$, and

$$f_3 - (x + 3)h = 4(x - 1).$$

All three original polynomials are divisible by $x - 1$, while the displayed steps put $x - 1$ in their ideal. Hence the ideal is $(x - 1)$ and a Groebner basis is $\{x - 1\}$.

Exercise 28.22: Euclidean algorithm giving the unit ideal

Write the two source generators as f, g. The successive remainder equations are

$$\begin{aligned} f &= (x^2+x)g + r_1, & r_1 &= x^2+3x-5,\\ g &= (x-4)r_1 + r_2, & r_2 &= 18x-21,\\ r_1 &= \left(\frac{x}{18}+\frac{25}{108}\right) r_2 - \frac{5}{36}. \end{aligned}$$

Each remainder lies in the ideal of the preceding pair, so the original ideal contains the nonzero constant $-5/36$, and therefore 1. The required Groebner basis is $\{1\}$.

Exercise 28.23: A two-point variety

Put $f = x^2y - x - 2$ and $g = xy + 2y - 9$. Then

$$f - (x-2)g = 4(2x + y - 5).$$

Thus $h = 2x+y-5$ lies in the ideal; furthermore $g-(y/2)h = -(y^2-9y+18)/2$. Set $k = y^2-9y+18$. These steps are reversible, so $(f, g) = (h, k)$. The leading monomials x and y^2 are relatively prime; the pair is a Groebner basis. Since $k = (y-3)(y-6)$, solve $y = 3$ or 6 and $x = (5-y)/2$. The real variety is $\{(1, 3), (-1/2, 6)\}$.

Exercise 28.24: A variety consisting of the origin

Let $f = x^2y + x$ and $g = xy^2 - y$. Their leading-term cancellation gives $yf - xg = 2xy$, so $h = xy$ lies in the ideal. Then $f - xh = x$ and $g - yh = -y$. Hence (f, g) contains both x, y. Conversely both f, g belong to (x, y), so the ideals are equal. Thus $\{x, y\}$ is a Groebner basis, and its common zero set is exactly $\{(0, 0)\}$.

Exercise 28.25: A cubic curve intersection reduced to one variable

For $f = x^2y + x + 1$ and $g = xy^2 + y - 1$, compute $yf - xg = x + y = h$. Dividing by h gives

$$g = y^2h - k, \qquad f = (xy - y^2 + 1)h + k, \qquad k = y^3 - y + 1.$$

These identities show $(f, g) = (h, k)$. The leading monomials x and y^3 are relatively prime, so $\{x + y, y^3 - y + 1\}$ is a Groebner basis. The real variety is $\{(-t, t) : t^3 - t + 1 = 0\}$. This cubic has exactly one real zero: its local maximum and minimum, at $t = \mp 1/\sqrt{3}$, are both positive, while its value tends to $-\infty$ as $t \to -\infty$. Thus the description gives a single real point without requiring a radical expression.

Exercise 28.26: A line and an isolated point

Let $f = x^2y + xy^2$ and $g = xy - x$. Set $h = f - (x + y + 1)g = x^2 + x$. Thus $(f, g) = (h, g)$. For lex order, their leading monomials are x^2 and xy. The only S-polynomial is

$$yh - xg = x^2 + xy = h + g,$$

which reduces to zero. Hence $\{x^2 + x, xy - x\}$ is a Groebner basis. The equations are $x(x + 1) = 0$ and $x(y - 1) = 0$. For $x = 0$, any y works; for $x = -1$, we must have $y = 1$. The variety is the entire y-axis together with the point $(-1, 1)$.

Exercise 28.27: True and false statements

(a) True. Here the book uses $F[\mathbf{x}] = F[x_1, \ldots, x_n]$. The Hilbert Basis Theorem gives a finite generating set for every ideal. This is not a claim that a multivariable ideal is principal.

(b) False. For example $\mathbb{R}^2 \setminus \{(0, 0)\}$ is not the zero set of polynomials: zero sets are closed, whereas this set is not.

(c) True. It is the zero set of the constant polynomial 1.

(d) True. A point (a, b) is the zero set of $(x - a)^2 + (y - b)^2$; a finite union of points is the zero set of the product of these polynomials.

(e) True. A line is the zero set of $ax + by + c$ with $(a, b) \neq (0, 0)$.

(f) True. A finite union of lines is the zero set of the product of their defining linear polynomials.

(g) True. The Euclidean algorithm preserves common divisors at each remainder step and terminates as degrees decrease.

(h) This depends on the reader's experience. If one has used the Euclidean algorithm in one variable or row reduction for linear polynomials, one has already computed special cases of Groebner bases; it is not a universal mathematical assertion about every reader.

(i) False. For example $\{x\}$ and $\{2x\}$ are distinct Groebner bases of (x). A *reduced monic* Groebner basis for a fixed monomial order is unique.

(j) False. $x \in (x, y)$ but $x \notin (x^2, y^2)$, since the latter ideal has no terms of degree one. Equal zero sets need not give equal ideals.

Exercise 28.28: Order is not divisibility

With $y < x$ in lex order, $y < x$ because the x exponent of y is zero and that of x is one. Yet y does not divide x: a product of y with a monomial must still have positive y exponent. This is the required counterexample.

Exercise 28.29: The ideal generated by a finite list

Source correction. The displayed coefficient condition $c_i \in I$ is circular; the intended condition is $c_i \in R$. With that correction define $I = \{\sum_{i=1}^{r} c_i f_i : c_i \in R\}$. Zero is obtained by all coefficients zero. A difference is obtained by subtracting the coefficients, so I is an additive subgroup. For $a \in R$,

$$a \sum_i c_i f_i = \sum_i (a c_i) f_i \in I.$$

Commutativity gives absorption on the right as well, proving the ideal claim. Unity ensures each f_i belongs to I, so this is exactly the smallest ideal containing the list.

Exercise 28.30: Common divisors survive a remainder step

If $d \mid f$ and $d \mid g$, then $d \mid (f - gq) = r$. Conversely if $d \mid g$ and $d \mid r$, then $d \mid (gq + r) = f$. Thus the common divisors of f, g are exactly those of g, r. This elementary equivalence is the reason the Euclidean algorithm computes a gcd.

Exercise 28.31: Checking a simple Groebner basis

For $f = xy$ and $g = y^2 - y$, the leading monomials under lex order $y < x$ are xy and y^2. Their least common multiple is xy^2, and

$$S(f, g) = yf - xg = xy = f.$$

It reduces to zero by subtracting f. Buchberger's criterion therefore proves that $\{xy, y^2 - y\}$ is a Groebner basis of the ideal it generates.

Exercise 28.32: The vanishing ideal of a set

The zero polynomial vanishes on S. If f, g vanish on S, then $(f - g)(s) = 0$ for every $s \in S$. If h is any polynomial, $(hf)(s) = h(s)f(s) = 0$ for every $s \in S$. These are precisely closure under differences and absorption, so $I(S)$ is an ideal in $F[x_1, \ldots, x_n]$.

Exercise 28.33: A set lies in its double construction

Take $s \in S$. By definition every $f \in I(S)$ vanishes at s. Hence s is a common zero of all polynomials in $I(S)$, which says $s \in V(I(S))$. Since s was arbitrary, $S \subset V(I(S))$.

Exercise 28.34: The double construction can enlarge a set

Take $S = \{(t, 0) : t \in \mathbb{R},\ t \neq 0\}$. If f vanishes on S, the one-variable polynomial $f(x, 0)$ has infinitely many roots and is therefore zero. Thus f is divisible by y, so $I(S) = (y)$. Consequently $V(I(S))$ is the full x-axis and contains $(0, 0)$, which is absent from S.

Exercise 28.35: An ideal lies in its double construction

If $f \in N$ and $s \in V(N)$, the definition of $V(N)$ gives $f(s) = 0$. Therefore f vanishes on every point of $V(N)$ and lies in $I(V(N))$. Thus $N \subset I(V(N))$. If $V(N)$ is empty, every polynomial vacuously vanishes on it and the inclusion still holds.

Exercise 28.36: The double construction can enlarge an ideal

Choose $N = (x^2, y^2) \subset \mathbb{R}[x, y]$. Its zero set is $\{(0, 0)\}$, so x vanishes on $V(N)$ and belongs to $I(V(N))$. But $x \notin N$: every term of every polynomial $ax^2 + by^2$ has total degree at least two. Hence $I(V(N)) \neq N$. In fact $I(V(N)) = (x, y)$.